

ROADMAP — File by FIFO (HR Disciplinary System)

A single place that records **what's been built** over ~6 months of development and **what's left to do**, priority-ordered. Synthesised from `RECENT_UPDATES.md`, `SESSION_HISTORY.md`, `PRE_LAUNCH_AUDIT_2026-06.md`, `legal/BUSINESS_LAUNCH_TRACKER.md`, the future-feature roadmap docs, and a direct read of the live codebase.

Last updated: 2026-06-04 · **Status:** Production-ready, deployed at file.fifo.systems, **0 paying clients yet**

Where we are now (baseline)

- **Frontend:** React 18 + Vite + TypeScript (strict) + Tailwind. Live at <https://file.fifo.systems>.
- **Backend:** Firebase Cloud Functions — Node 22, `firebase-functions` v7, `firebase-admin` v13, **all on the v2 API** (~34 functions). Primary region `us-central1`, super-user functions `us-east1`.
- **Data:** Multi-tenant **sharded** Firestore — every tenant's data under `organizations/{orgId}/...`, isolation enforced by security rules. Architected for 2,700+ orgs.
- **Email:** SendGrid from `file@fifo.systems` (App Emails). Payment gateway not yet integrated.
- The product is feature-complete and hardened; the gap to revenue is **commercial/business setup**, not engineering.

✓ What's been done (retrospective, by epic)

Warning workflow

- Full **V2 warning wizard rewrite** — intro overview + 5 working steps (Setup → Incident → Conversation → Sign & Save → Deliver). **V1 (10-phase) deleted 2026-06-03** — V2 is the sole wizard.
- LRA-compliant **escalation engine** (counselling → verbal → first/second/final written) with real-time recommendation and category override.
- **HR-intervention gate** prevents auto-escalation past final written; urgent HR alerts on repeat offences.
- **Multi-manager** system (`managerIds[]`) with bulk assignment, backward-compatible helpers.

PDF system (3-layer, legal-grade)

- **Versioned generator** — v1.0.0 [frozen], v1.1.0, v1.2.0 [template-aware]; historical warnings regenerate byte-identically for CCMA admissibility.
- **Per-org template customisation** (colours, fonts, margins, section order/visibility) + **centralised version storage** (~1000× storage reduction — 5-byte reference vs 5-10 KB object per warning).
- CCMA-ready output: lettered sections, continuation headers, electronic-signature notation, page initials, **Employee Rights** block, watermarked **appeal-decision reports**.
- **SVG signatures** (2-5 KB vs 50-200 KB PNG, ~90% smaller) with witness watermarking.

Employee / manager / department management

- Full CRUD; **CSV import** with SA phone (`082...` → `+2782...`) and date formatting; duplicate detection.
- **Dual-mode user creation** (promote existing employee or create new manager); custom claims set immediately (no sign-out/in cycle).
- **Department system** with real-time counts + default departments.
- **Manual/historical warning entry** for digitising paper records, with a **60-day countdown** to encourage digitisation.

Delivery — all four methods live

- **Email, QR** (1-hr signed link), **WhatsApp** (durable 180-day respond link via `wa.me`), **Printed** (collect-from-HR with `notifyHRPrintedCollection`).
- Fixed the `deliveryStatus` -vs- `status` bug that left delivered warnings stuck in the undelivered queue.

Employee response & appeal

- Public `/respond/:token` page (no auth), **8 Cloud Functions**, token auth + rate limiting, PDF viewing, evidence upload, and **HR email notifications** on response/appeal via SendGrid.

Security & architecture

- Database **sharding** for scale; **server-side timestamps** (tamper-proof audit trail); **session guard** (10-min inactivity logout) + forced app-update mechanism.
- **Evidence-upload hardening** (download-token URLs not public; client-side image optimisation).
- **Reseller demo organisations** with persistent banner + cron/trigger guards (no fake-email bounces).
- **Tier 0 pre-launch hardening done (2026-06-04)**: `TEMP_LINK_SECRET` moved to `defineString` (fails closed, no hardcoded fallback); `getHREmails()` repointed to the sharded `organizations/{orgId}/users` path across `warningDelivery.ts` , `employeeResponse.ts` , `notifyHROnAppeal.ts` .

Design / UX / accessibility

- Unified design system; **WCAG 2.1 AA** modals (focus trap, scroll lock, standardised z-index 9000-9999); mobile polish; **multi-dashboard theming** (Manager/HR/Executive per-view colours); signature-pad redesign.

Infrastructure

- **Node 20 → 22** + full **v1 → v2 functions migration**; build/version pipeline with forced updates; performance + **staleness-detection** optimisations; production console-log scrub (`no-console` enforced).

Business / legal / marketing

- **FIFO Solutions (Pty) Ltd registered** (2026/071559/07); legal docs (ToS, Privacy, Reseller Agreement); full sales/marketing toolkit; **HR-consultant-as-reseller** go-to-market validated.

Large-scale cleanup (Phases 0-6)

- tsc errors 903 → 647; ~10,000 LOC removed; `DataService` → `ShardedDataService` + `AdminDataService`; PDF generator 4,135 → 958 LOC (byte-identical); orphan-record audit + recurrence patches; lint clean.

● What's left — priority-ordered

P0 — Blocks the first paying client (commercial/revenue)

The *engineering* pre-launch blockers (audit Tier 0) are **done**. What remains is commercial setup: - **SARS registration** — tax number needed before you can invoice clients (TIN submitted). - **Capitec business account** — finalise (needed to receive payment). - **Payment collection** — no gateway integrated yet (PayFast/Yoco). Needed before self-serve billing; first clients can be invoiced manually in the interim. (`CommissionService.ts:617` commission-statement PDF gen is still a `TODO`.) - **Sign the reseller agreement** with Adam. - **Onboard the first client** — Insitu Construction (Michelle Pedersen) is closest; needs an onsite visit to capture the staff list and set up the company profile.

P1 — During the first-50 ramp

- **End-to-end test the "done, needs testing" features** (built but never verified end-to-end):
 - Gap 4 — manager evidence upload in the wizard (attach → complete → confirm in Storage/Firestore).
 - Gap 5 — expected-standards templates auto-fill on category select.
 - Response-link flow — generate → view PDF → submit response/appeal → HR notification.
 - Historical-warning 60-day countdown; department real-time counts; HOD team view; employee edit after promotion.

- **Mobile-first pass on the hot paths** (the honest “responsive  , mobile-first × ” gap). Drive the warning wizard, employee list, HOD dashboard and login at **320px on a real low-end Android / 3G throttle**; fix tap targets and horizontal scroll. Delete the dead, unimported `MobileEmployeeManagement.tsx` (~335 lines).
- **Pricing review with Adam** — Tiers 4-6 are well below market (a 200-emp client at R6.25/emp/mo is ~77% cheaper than Sage HR). Consider a “2026 pricing” tier for new clients; grandfather early ones. Estimated upside ≈ **+R360k/yr** at 300 clients. See `legal/BUSINESS_LAUNCH_TRACKER.md`.
- **Research only (no implementation)**: unified warning/counselling architecture analysis — should the corrective-counselling sections live *inside* the warning (per the client template), and at what migration cost?

P2 — Scale-prep (sequence past ~300 orgs; do NOT do prematurely at 0 clients)

- **Cron read amplification** — `checkDueReviewsDaily` and `cleanupExpiredAudio` both loop all orgs (~10.8k reads/day at 2,700 orgs). Stagger schedules + batch/paginate org processing.
- **Firestore index drift** — ~14 production indexes exist only in the Firebase console, not in `config/firestore.indexes.json`. Export and reconcile into version control (do once).
- **Function-level content validation** — Admin SDK writes bypass security rules; add a guard (`if (data.organizationId !== orgIdFromPath) throw`) as cheap insurance.
- **CI hardening** — add a Playwright **multi-tenant isolation** test (org A can’t see org B’s warning) + daily prod smoke test + Dependabot/Snyk.
- **Automate backups + write a one-page DR runbook** (backup script exists but is manual).
- **Lower-priority debt**: Firestore listener pooling/dedup (~25 `onSnapshot` , measure first); decompose god components opportunistically (`UnifiedWarningWizardV2` , `ClientOrganizationManager` , `EnhancedOrganizationWizard`); bundle weight (revisit only on 3G cold-load complaints).

P3 — Future features (design docs already written)

- **Review Follow-Up System** — track warnings post-issuance, auto-satisfy/escalate. 4-week plan in `REVIEW_FOLLOWUP_IMPLEMENTATION_ROADMAP.md`.
- **Recognition & Achievement Tracking** — positive-reinforcement counterpart to discipline. 4-6 week plan in `RECOGNITION_SYSTEM_IMPLEMENTATION_ROADMAP.md`.

In-code debt (small, opportunistic)

- `frontend/src/types/core.ts:512-518` — 4 `@deprecated` alias fields (`lastName` , `department` , `position` , `category`) to retire.
- `frontend/src/types/core.ts:537` — `FIXME` : reconcile `actionSteps` {`action`,`timeline`} shape (Phase 2/3 cleanup).
- `frontend/src/services/ErrorTrackingService.ts:165` — `TODO` : wire external error tracking.

- `frontend/src/services/CommissionService.ts:617` — `TODO`: commission-statement PDF generation.
-

Where the detail lives

- `RECENT_UPDATES.md` — Sessions 20-67 change log (current)
- `SESSION_HISTORY.md` — Sessions 5-19 (archived)
- `PRE_LAUNCH_AUDIT_2026-06.md` — Tier 0/1/2 audit (note: Tier 0 now done; Tier 1.3 dual-wizard item resolved)
- `legal/BUSINESS_LAUNCH_TRACKER.md` — company formation, banking, pricing, go-to-market
- `REVIEW_FOLLOWUP_IMPLEMENTATION_ROADMAP.md` / `RECOGNITION_SYSTEM_IMPLEMENTATION_ROADMAP.md` — P3 designs
- `CLAUDE.md` + `QUICK_REFERENCE.md` — working guidance and file catalog